

Unit 12, Lesson 2: Using Proportions to Solve Problems



Benchmark

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.



Learning Target

☐ I can solve problems using proportions.



12.2.1 Warm-Up: Making More Servings

1. Gabriel wants to make 5 servings of oatmeal for breakfast using this recipe. How much water and oats will he need to use? Show your work.

TRADITIONAL STEEL CUT OATS

Hot Water or Milk Preparation

1 Serving

- 1-1/2 cups water or milk
- 1/4 cup Quaker Steel Cut Oats

Directions:

1. Bring water or milk to a boil in a medium saucepan.
2. Stir in oats, reduce heat to low.
3. Simmer uncovered over low heat, stirring occasionally, for 25-30 minutes or until oats are of desired texture.



12.2.2 Guided Instruction: Solving Proportions

Just as there are different ways to set up a proportion to solve a problem, there are different ways to solve the proportion once it's set up.

Jacksonville and Gainesville are approximately 1.4 inches (in.) apart on a scaled map of Florida. The scale used for the map is 2 in. = 90 miles. The proportion below can be used to find the actual distance between Jacksonville and Gainesville.

$$\frac{2}{90} = \frac{1.4}{x}$$

Two methods for solving the proportion are shown.

Method A Using a Scale Factor	Method B Solving the Equation
$\frac{2}{90} = \frac{1.4}{x}$ Determine the factor multiplied by 2 to get 1.4. $2 \div 1.4 = 0.7$ $\frac{2}{90} \times \frac{0.7}{0.7} = \frac{1.4}{63}$	$\frac{2}{90} = \frac{1.4}{x}$ $\cancel{90}^1 \cdot \frac{2}{\cancel{90}_1} = \frac{1.4}{x} \cdot 90$ $x \cdot 2 = \frac{126}{x} \cdot x^1$ $\frac{12x}{2 \cdot 1} = \frac{126}{2}$ $x = 63$
Both solution methods show the approximate distance between Jacksonville and Gainesville is 63 miles.	

1. In Method A, how does multiplying the ratio $\frac{2}{90}$ by $\frac{0.7}{0.7}$ show that $\frac{2}{90}$ and $\frac{1.4}{63}$ is equivalent?
2. In Method B, why were both ratios multiplied by 90 and then by x ?
3. In Method B, why are both sides of the equation multiplied or divided by the same value in each step?

A certain shade of light blue paint is made by mixing $2\frac{1}{2}$ quarts of blue paint with 7 quarts of white paint. To make the same shade of light blue, how much white paint would be needed to mix with 4 quarts of blue paint?

4. Complete the proportion using the information given.

$$\frac{2.5}{\underline{\quad}} = \frac{7}{\underline{\quad}}$$

5. With your partner, decide who is Partner A and who is Partner B. Then solve the proportion using the method indicated in the space provided.

Partner A - Method A Using a Scale Factor	Partner B - Method B Solving the Equation

6. Check your work with your partner, making any necessary corrections. Once both partners agree, write your partner's method in your workbook. Then complete the statement.

To make the same shade of light blue, _____ quarts of white paint would be needed.



12.2.3 Collaboration: Examining Student Work

Four students solved different problems involving proportional relationships. Each student's work is shown, including any errors they may have made. Work with your partner or a small group to examine each student's work and complete the table that follows.

Student	Javier
Problem	A mixture of concrete is made up of sand and cement in a ratio of 5: 3. How many cubic feet of sand is needed to make 160 cubic feet of concrete?
Student's Work	

What did Javier do correctly?	What error(s) did Javier make?	Find the correct answer.



Student	Kayla
Problem	A recipe for making 4 pancakes calls for the ingredients listed. ☐ 6 tablespoons flour ☐ $\frac{1}{4}$ pint milk ☐ 1 pinch salt ☐ 1 egg ☐ $\frac{1}{3}$ pint water You want to make 10 pancakes. - How much flour do you need? - How much milk do you need?
Student's Work	a. How much flour do you need? $10 : 4 = ? : 6$ $\text{Flour} = \frac{10}{4} = \frac{?}{6}$ $10 \text{ pancakes} = 6 + 6 + 3$ $= 15 \text{ spoonfuls of flour}$ b. How much milk do you need? $10 \text{ pancakes} = \frac{1}{4} + \frac{1}{4} + \frac{1}{8} = \frac{3}{16} \text{ pints}$

What did Kayla do correctly?	What error(s) did Kayla make?	Find the correct answers.

Student	Isaac
Problem	A drawing of a surfboard in a catalog shows its length as $8\frac{4}{9}$ inches (in.). Find the actual length of the surfboard if $\frac{1}{2}$ in. length on the drawing corresponds to $\frac{3}{8}$ foot of actual length.
Student's Work	$\frac{\frac{1}{2}}{8\frac{4}{9}} = \frac{\frac{3}{8}}{5}$ $\cancel{8.44} \cdot \frac{0.5}{8.44} = \frac{0.38}{5} \cdot 8.44$ $5 \cdot 0.5 = \frac{3.2072}{5} \cdot 8$ $\frac{0.55}{0.5} = \frac{3.2072}{0.5}$ $5 = 6.4 \text{ in.}$

What did Isaac do correctly?	What error(s) did Isaac make?	Find the correct answers.

Student	Mari
Problem	Juan made 13 out of 20 free throws at basketball practice. If Jada shoots 25 free throws, what is the smallest number of shots she has to make to have a better free-throw percentage than Juan?
Student's Work	$\frac{13}{20} = \frac{f}{25}$ $20 + 5 = 25$ $13 + 5 = 18$ Jada has to make 18 shots.

What did Mari do correctly?	What error(s) did Mari make?	Find the correct answers.



12.2.4 Bring It Together: Proportions

1. Mistakes create opportunities to learn! Based on the mistakes you observed when examining other students' work, use the space provided to describe at least two things you learned and want to keep in mind when using proportions to solve problems.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills



12.2.5 Your Turn!

Solve each problem using proportional reasoning. Show your work.

1. Tyler swims at a constant speed of 5 meters (m) every 3 seconds. How long does it take him to swim 114 meters?
2. To make a shade of paint called jasper green, mix 4 quarts (qt) of green paint with $\frac{2}{3}$ cup (c) of black paint. How much green paint should be mixed with 4.5 cups of black paint to make jasper green?
3. It takes an ant farm 3 days to consume $\frac{1}{2}$ of an apple. At that rate, in how many days will the ant farm consume 6 apples?



12.2.6 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate where your current learning is in relation to today's learning target.

I can solve problems using proportions.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



2. Explain why you placed your **X** where you did.